

$$\begin{aligned}
r_{k_i} &= y_k - h_k(\hat{x}_{k_i}^-, 0) \\
S_{k_i} &= H_{k_i} P_{k_i}^- H_{k_i}^T + R_k \\
\beta_{k_i} &= \frac{\exp[-r_{k_i}^T S_{k_i}^{-1} r_{k_i} / 2]}{(2\pi)^{n/2} |S_{k_i}|^{1/2}} \\
a_{k_i} &= \frac{a_{k-1,i} \beta_{k_i}}{\sum_{j=1}^M a_{k-1,j} \beta_{k_j}} \quad (13.107)
\end{aligned}$$

Note that the weighting coefficient a_{k_i} is computed by using the measurement y_k to obtain the relative confidence β_{k_i} of the estimate $\hat{x}_{k_i}^-$. The pdf of the *a posteriori* state estimate is obtained by the following sum:

$$\text{pdf}(\hat{x}_k^+) = \sum_{i=1}^M a_{k_i} N(\hat{x}_{k_i}^+, P_{k_i}^+) \quad (13.108)$$

This approach can also be extended to smoothing [Kit94]. Similar approaches can be taken to expand the pdf using non-Gaussian functions [Aok67, Sor68, Sri70, deF71, Hec71, Hec73, Mcr75, Wil81, Kit87, Kra88]. A related filter has been derived for the case where either the process noise or the measurement noise is strictly Gaussian, but the other noise is Gaussian with heavy tails [Mas75, Tsa83]. This is motivated by the observation that many instances of noise in nature have pdfs that are approximately Gaussian but with heavier tails [Mas77].

Another approach to nonlinear filtering is called grid-based filtering. In grid-based filtering, the value of the pdf of the state is approximated, stored, propagated, and updated at discrete points in state space [Buc69, Buc71]; [Spa88, Chapter 6]. This is similar to particle filtering (discussed in Chapter 15), except in particle filtering we choose the particles to be distributed in state space according to the pdf of the state. Grid-based filtering does not distribute the particles in this way, and hence has computational requirements that increase exponentially with the dimension of the state. Grid-based filtering is even more computationally expensive than particle filtering, and this has limited its application. Furthermore, particle filtering is a type of “intelligent” grid-based filtering. This seems to portend very little further work in grid-based filtering.

Richard Bucy suggested yet another approach to nonlinear filtering [Buc65]. Instead of linearizing the system dynamics, compute the theoretically optimal nonlinear filter, and then linearize the nonlinear filter. However, the theoretically optimal nonlinear filter is very difficult to compute except in special cases.

13.4 PARAMETER ESTIMATION

State estimation theory can be used to not only estimate the states of a system, but also to estimate the unknown parameters of a system. This may have first been suggested in [Kop63]. Suppose that we have a discrete-time system model, but the system matrices depend in a nonlinear way on an unknown parameter vector p :

$$\begin{aligned}
 x_{k+1} &= F_k(p)x_k + G_k(p)u_k + L_k(p)w_k \\
 y_k &= H_k x_k + v_k \\
 p &= \text{unknown parameter vector}
 \end{aligned} \tag{13.109}$$

In this model, we are assuming that the measurement is independent of p , but this is only for notational convenience. The discussion here can easily be extended to include a dependence of y_k on p . Assume that p is a constant parameter vector. We do not really care about estimating the state, but we are interested in estimating p . This is the case, for example, in the aircraft engine health estimation problem [Kob03, Sim05a]. In those papers it was assumed that we want to estimate aircraft engine health (for the purpose of maintenance scheduling), but we do not really care about estimating the states of the engine.

In order to estimate the parameter p , we first augment the state with the parameter to obtain an augmented state vector x' :

$$x'_k = \begin{bmatrix} x_k \\ p_k \end{bmatrix} \tag{13.110}$$

If p_k is constant then we model $p_{k+1} = p_k + w_{pk}$, where w_{pk} is a small artificial noise term that allows the Kalman filter to change its estimate of p_k . Our augmented system model can be written as

$$\begin{aligned}
 x'_{k+1} &= \begin{bmatrix} F_k(p_k)x_k + G_k(p_k)u_k + L_k(p_k)w_k \\ p_k + w_{pk} \end{bmatrix} \\
 &= f(x'_k, u_k, w_k, w_{pk}) \\
 y_k &= \begin{bmatrix} H_k & 0 \end{bmatrix} \begin{bmatrix} x_k \\ p_k \end{bmatrix} + v_k
 \end{aligned} \tag{13.111}$$

Note that $f(x'_k, u_k, w_k, w_{pk})$ is a nonlinear function of the augmented state x'_k . We can therefore use an extended Kalman filter (or any other nonlinear filter) to estimate x'_k .

■ EXAMPLE 13.4

This example is taken from [Ste94]. Suppose we have a second-order system governed by the following equations:

$$\ddot{x}_1 + 2\zeta\omega_n\dot{x}_1 + \omega_n^2 x_1 = \omega_n^2 w \tag{13.112}$$

where ω_n is the natural frequency of the system, ζ is the damping ratio, and the input w is zero-mean noise. A state-space model for this system can be written as

$$\begin{aligned}
 \dot{x}_1 &= x_2 \\
 \dot{x}_2 &= -\omega_n^2 x_1 - 2\zeta\omega_n x_2 + \omega_n^2 w \\
 \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} w
 \end{aligned} \tag{13.113}$$

Suppose that $-2\zeta\omega_n$ is known, but ζ and ω_n are unknown. We want to estimate $-\omega_n^2$. Suppose that both x_1 and x_2 are available for measurement. We define the known parameter as b ; that is, $b = -2\zeta\omega_n$. We define a new state element equal to the parameter that we want to estimate. That is, $x_3 = -\omega_n^2$. We then form an augmented system model as follows:

$$\begin{aligned}\dot{x}' &= \begin{bmatrix} x_2 \\ x_3x_1 + bx_2 - x_3w \\ w_p \end{bmatrix} \\ &= f(x', w') \\ w' &= \begin{bmatrix} w \\ w_p \end{bmatrix} \\ y &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} x' + v\end{aligned}\quad (13.114)$$

where w_p is an artificial noise term that we add to the system that allows the Kalman filter to modify its estimate of x_3 . We can use an extended Kalman filter to estimate the augmented state. First we need to find the partial derivative matrices:

$$\begin{aligned}F &= \left. \frac{\partial f}{\partial x'} \right|_{\hat{x}', w'_0} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ x_3 & b & x_1 - w \\ 0 & 0 & 0 \end{bmatrix}_{\hat{x}', w'_0} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ \hat{x}_3 & b & \hat{x}_1 \\ 0 & 0 & 0 \end{bmatrix} \\ L &= \left. \frac{\partial f}{\partial w'} \right|_{\hat{x}', w'_0} \\ &= \begin{bmatrix} 0 & 0 \\ -\hat{x}_3 & 0 \\ 0 & 1 \end{bmatrix}\end{aligned}\quad (13.115)$$

The continuous-time extended Kalman filter can be written as

$$\begin{aligned}\dot{\hat{x}}' &= f(\hat{x}', 0) + K(y - H\hat{x}') \\ K &= PH^TR^{-1} \\ \dot{P} &= FP + PF^T + LQL^T - PH^TR^{-1}HP\end{aligned}\quad (13.116)$$

Figure 13.4 illustrates the results of a typical simulation of the extended Kalman filter that is used to estimate $-\omega_n^2$ for this system. The true system parameters are $\omega_n = 2$ and $\zeta = 0.1$, so $-\omega_n^2 = -4$. Suppose that we begin by estimating $-\omega_n^2$ as -8 with an initial estimation variance of 20. Figure 13.4 shows that the error in our estimate of $-\omega_n^2$ gradually decreases toward zero, and the estimation variance gradually decreases. We set the variance of the artificial noise w_p equal to 0.1 in this example. This allows the Kalman filter